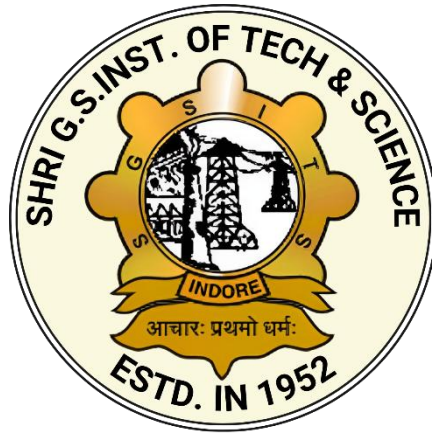


Smart Robotic Arm for Automated Material Sorting in Manufacturing



SESSION: Jan 2026- June 2026

A Minor Project Report

Submitted to

Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal towards the
partial fulfilment of the Degree of
Bachelor of Technology

Electronics and Telecommunication Engineering

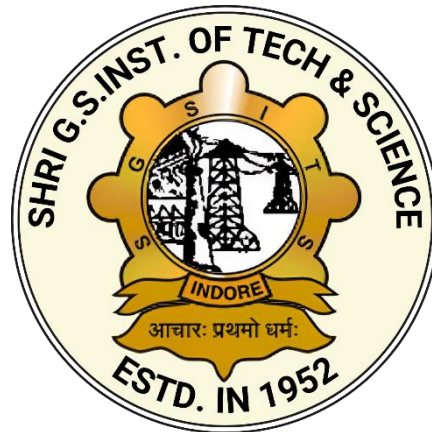
Guided by:

Prof. Shekhar Sharma
Prof. Neeraj Malviya

Submitted by:

Akshat Rangwani (0801EC231014)
Ankit Chaurasiya (0801EC231022)
Lakshya Garge (0801EC231056)
Somil Agrawal (0801EC231098)

SHRI GOVINDRAM SEKASARIA INSTITUTE OF TECHNOLOGY AND
SCIENCE, INDORE (M.P.)



RECOMMENDATION

This major project report entitled, "**Smart Robotic for automated material sorting in Manufacturing**", submitted to the Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal, by Akshat Rangwani, Ankit Chaurasiya, Lakshya Garge, Somil Agrawal during the academic year 2026 as a partial fulfilment for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering, is a record of student's own work carried out by them under our direct supervision, in the Department of Electronics and Telecommunication Engineering, S.G.S.I.T.S. Indore (M.P.). The work contained in the project report is a satisfactory account of their project work and is recommended for the award of the degree.

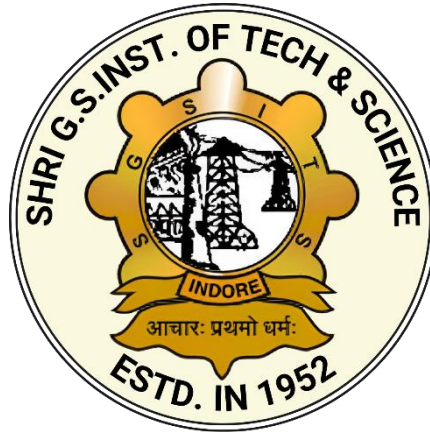
Guide Name
Prof. Shekhar Sharma

Co-guide Name
Prof. Neeraj Malviya

Prof. S. K. Jain
(Head of Department)

SHRI GOVINDRAM SEKASARIA INSTITUTE OF TECHNOLOGY AND SCIENCE,
INDORE (M.P.)
Electronics and Telecommunication Engineering

CERTIFICATE



This is to certify that the minor project report entitled " Smart Robotic for automated material sorting", submitted to the Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal, by Akshat Rangwani, Ankit Chaurasiya, Lakshya Garge, Somil Agrawal during the academic year 2025-2026, is a record of their own work and is accepted in the partial fulfilment for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering.

Internal Examiner:

External Examiner:

Date:

Date:

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who contributed to the successful completion of this minor project. First and foremost, we are deeply thankful to our project guide, Prof. Neeraj Malviya and Prof. Shekhar Sharma, for their valuable guidance, continuous encouragement, and constructive suggestions throughout the project.

We extend our heartfelt thanks to Prof. S. K. Jain, Head of the Department of Electronics and Telecommunication Engineering, for providing all the necessary facilities and a conducive environment to carry out this work.

We are also grateful to the faculty members of our department, our classmates, and our families for their constant support and motivation. The technical resources, open-source communities (Arduino IDE, OpenCV), and the 3D printing lab staff were indispensable to this project.

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Abstract

In modern industrial and academic environments, the need for automated object handling and sorting systems has grown rapidly. This minor project presents the design, fabrication, and programming of a material-based waste segregation system using a 3D-printed robotic arm controlled by an ESP32 microcontroller, integrated with computer vision pipeline.

The robotic arm is constructed using STL (Standard Tessellation Language) files sourced from open-source repositories such as Thingiverse and printed using PLA filament. It is actuated by six servo motors that provide three degrees of freedom and a gripping mechanism. Object detection and material classification are achieved using an OpenCV-based Python script running on a host computer, which identifies waste material objects placed on a conveyor-like platform and sends serial commands to the Arduino.

The Arduino receives material-class codes (Organic, Recyclable) and executes precise servo motor sequences to pick the object and drop it into the designated bin. Testing demonstrated a classification accuracy of approximately 90% under consistent lighting conditions, with a cycle time of 4–6 seconds per object. This project demonstrates a cost-effective, scalable approach to industrial sorting automation, suitable for small-scale manufacturing, waste segregation, and educational robotics labs.

Chapter 1: Introduction

1.1 Background and Motivation

Automation has become the backbone of modern manufacturing. Sorting and segregation tasks—traditionally performed by human workers—are repetitive, time-consuming, and prone to error. Robotic arms can outperform humans in speed, accuracy, and consistency on such tasks.

The integration of 3D printing with robotics has dramatically lowered the entry barrier for building custom robotic manipulators. Open-source hardware platforms like ESP32, combined with open-source vision libraries such as OpenCV, allow engineering students to prototype fully functional industrial-grade systems on minimal budgets. This project leverages these technologies to build an automated waste-segregating robotic arm that can be replicated, modified, and deployed with ease.

1.2 Problem Statement

Manual sorting of objects on the basis of material type is tedious and inefficient in environments such as recycling plants, e-waste collection centres, and municipal waste facilities. There is a need for a low-cost, reliable automated system that can identify object material types in real time and mechanically sort them into designated categories.

1.3 Objectives

- Design and 3D print a robotic arm using open-source STL files.
- Integrate servo motors for controlled multi-axis movement.
- Implement a material detection algorithm using OpenCV (Python).
- Develop Arduino IDE to receive material codes and actuate the arm accordingly.
- Achieve reliable sorting of at least two waste categories (Organic, Recyclable).
- Validate the system with a prototype demonstration.

1.4 Scope of the Project

The scope is limited to a laboratory-scale prototype capable of sorting lightweight objects (up to 60 g) based on two waste material types under controlled lighting. The system is not intended for industrial heavy-load applications but serves as a proof-of-concept for scalable automation.

Chapter 2: Literature Review

2.1 Robotic Arm Automation

Robotic manipulators have been a subject of extensive research since the 1960s. Early industrial robots such as the UNIMATE were hydraulically actuated and reserved for heavy manufacturing. The advent of servo motor technology and microcontrollers enabled smaller, more precise desktop robotic arms. Researchers have proposed various kinematic models for calculating joint angles for desired end-effector positions—most commonly Denavit-Hartenberg (D-H) parameterisation. For this project, inverse kinematics is simplified using pre-calibrated servo angle lookup tables appropriate for the arm's fixed workspace.

2.2 Material Detection Technologies

A dominant approach exists for material detection in robotics:

Camera-based computer vision (OpenCV): A webcam captures image frames, which are processed in CNN material space. CNN-based segmentation is robust to illumination variation when ranges are properly calibrated. OpenCV's contour detection is then used to locate and classify objects.

2.3 3D Printed Robotic Arms

Open-source platforms like SolidWorks host numerous parametric robotic arm STL designs. PLA (Polylactic Acid) are the most commonly used filaments, with PLA preferred for its ease of printing and sufficient structural rigidity for lightweight payloads.

2.4 ESP32-Based Control Systems

Microcontrollers (ESP32) have been extensively used in academic robotic projects due to their open-source ecosystem, low cost, and wide community support. The Arduino Servo library facilitates PWM-based servo control, while the Serial library enables communication with a host computer running Python/OpenCV, creating a seamless hardware-software bridge.

Chapter 3: Components, Specifications and Dataset

3.1 Hardware Components

The following table lists all hardware components used in this project:

S.No.	Component	Model/Specification	Quantity	Purpose
1	Microcontroller	• ESP32 • Acts as central brain. • Processes sensor input data. • Sends signals to motors.	1	Main control unit
2	Servo Motors	• MG996R (Torque: 9.4 kg·cm) • Provide precise angular motion. • Control joints of arm. • Offer high torque output.	3	Arm joint actuation
3	Servo Motor (Gripper)	• SG90 (Torque: 1.8 kg·cm) • Manages the pinching action. • Regulates the gripping force. • Secures objects during movement.	3	Grip/release object
4	Webcam	• Captures live visual data. • Enables object recognition tasks. • Provides feedback for AI.	1	OpenCV vision input
5	3D Printed Arm Parts	• PLA • Form the physical skeleton.	1 set	Mechanical structure

S.No.	Component	Model/Specification	Quantity	Purpose
		• Lightweight but durable plastic. • Customized for specific reach.		
6	Power Supply	• 5V • Delivers steady electrical current. • Matches motor voltage needs. • Prevents system brownout issues.	1	Servo power
7	USB Cable	• Type-B • Connects board to computer. • Uploads the control code. • Provides basic serial communication.	1	Arduino programming
8	Jumper Wires	• Create flexible electrical paths. • Connect components without solder. • Link sensors to microcontroller.	50	Circuit connections

3.2 Software Tools

- Arduino IDE – Firmware development and serial communication
- Python with OpenCV (cv2), PyFirmata, NumPy – Computer vision
- Ultimaker Cura / PrusaSlicer – STL slicing for 3D printing

- SolidWorks – Open-source arm design
- VS Code – Python code editor

3.3 STL Files and 3D Printing Details

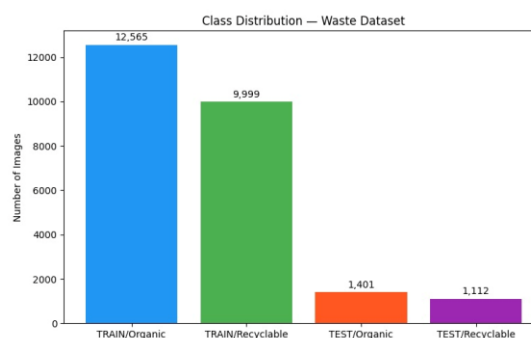
The EEZY bot ARM MK2 design was selected from Thingiverse (<https://www.thingiverse.com/thing:1454048>) due to its well-documented assembly guide, availability of all required STL parts, and suitability for MG996R servos. The following parts were printed:

1. Base plate – Houses the base servo and provides stability
2. Lower arm link – Connects base to elbow joint
3. Upper arm link – Connects elbow to wrist joint
4. Wrist assembly – Provides forward reach
5. Gripper finger pair (x2) – For object grasping
6. Servo horn adapters – Custom fit for MG996R

Parameter	Value
Filament	PLA 1.75 mm
Layer Height	0.2 mm

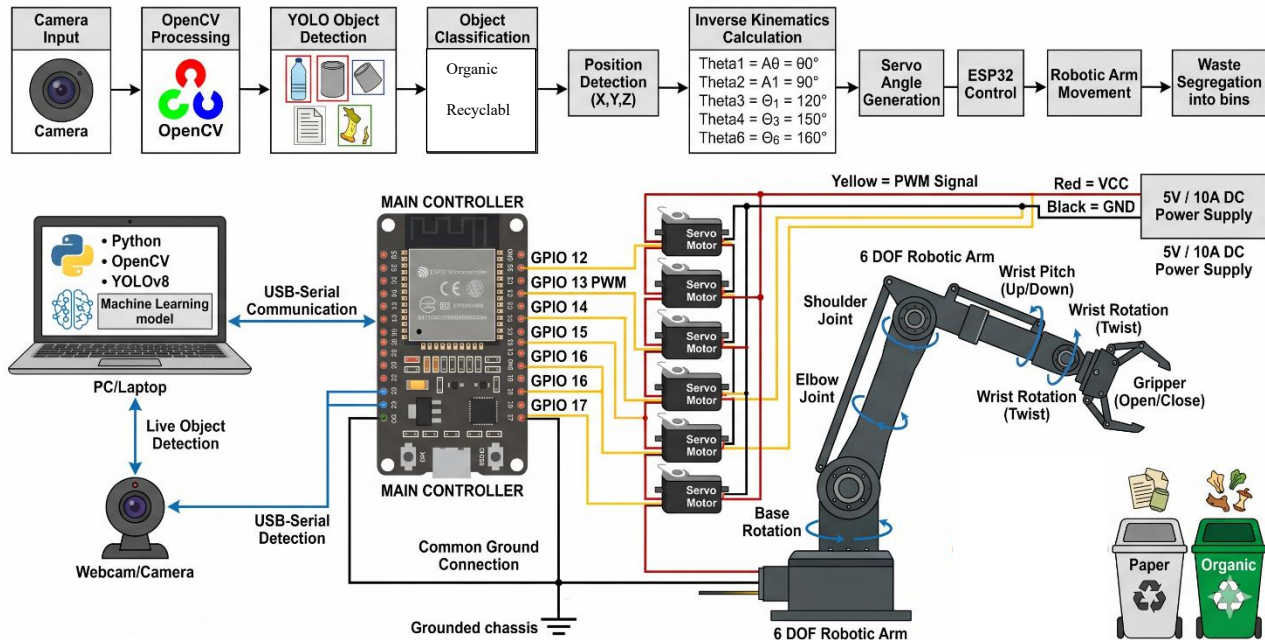
3.4 Dataset used

Material	Total Training
Organic	12565
Recyclable	9999



Chapter 4: System Design and Methodology

4.1 Diagram



4.2 Workflow

1. A material sample is placed on the detection platform in front of the webcam.
2. The Python script captures an image frame from the webcam and detect material using CNN.
3. The program analyzes the object's shape, size, and texture features (using open CV2) to identify the material type.
4. The material is classified into one of the predefined categories (organic, recyclable)
5. A single-character command is transmitted from Python to ESP32 using PySerial communication.
6. The ESP32 receives the command and executes the corresponding pick-and-place servo motor sequence using the MG996R motors.
7. The robotic arm picks up the detected material, rotates toward the assigned collection bin (organic, recyclable) places the material into the respective bin, and then returns to its home position.

4.3 Degrees of Freedom and Kinematics

The arm has 6 Degrees of Freedom (DOF):

- Joint 1 (Base rotation): Azimuthal rotation, range 0° – 180°
- Joint 2 (Shoulder): Elevation of lower arm, range 20° – 100°
- Joint 3 (Elbow): Elevation of upper arm, range 0° – 90°
- Gripper: Open/close, 40° (open) to 0° (closed)

Pre-calibrated angle sets are used for two bin positions (Left = Red, Right = Blue) and a fixed pick position above the detection platform.

4.4 Material Detection Algorithm

OpenCV processes each frame in the following steps:

1. Capture the live video frame from the camera and preprocess it for waste object detection.
2. Convert the captured frame into grayscale and apply Gaussian Blur (5×5 kernel) to reduce noise and improve detection accuracy.
3. Apply image thresholding and edge detection techniques to highlight the waste object from the background.
4. Detect contours in the processed frame to identify the object boundaries.
5. Select the largest contour above the predefined area threshold (>500 px) as the target waste object.
6. Extract object features such as shape, size, and texture, and classify the waste into categories (Organic, Recyclable) using the trained detection model.
7. Send the classified waste command to the ESP32 through serial communication for robotic arm movement.
8. Display the bounding box and waste category label on the output frame for real-time verification.

Chapter 5: Software Implementation

5.1 Arduino Code Structure

The Arduino firmware is structured as follows:

- Include the Servo library and define servo objects for all six motors.
- `setup()`: Attach servos to pins, move to home position, initialise Serial at 115200 baud rate.
- `loop()`: Read incoming serial byte; call the appropriate sorting function.
- `sortObject(char material)`: Executes pick, rotate, place, return sequence for the given material.
- Helper functions: `pickObject()`, `placeObject()`, `returnHome()`, `rotateTo(int angle)`.

Chapter 6: Results and Testing

6.1 Testing Methodology

Testing was conducted in two phases:

1. Unit testing: Individual servos tested for correct
2. Integration testing: Full pick-and-place cycle tested 50 times for each material.

6.2 Material Detection Accuracy

Material	Total Tests	Correct	Accuracy (%)
Organic	1401	1353	92%
Recyclable	1112	922	85%

6.3 Performance Metrics

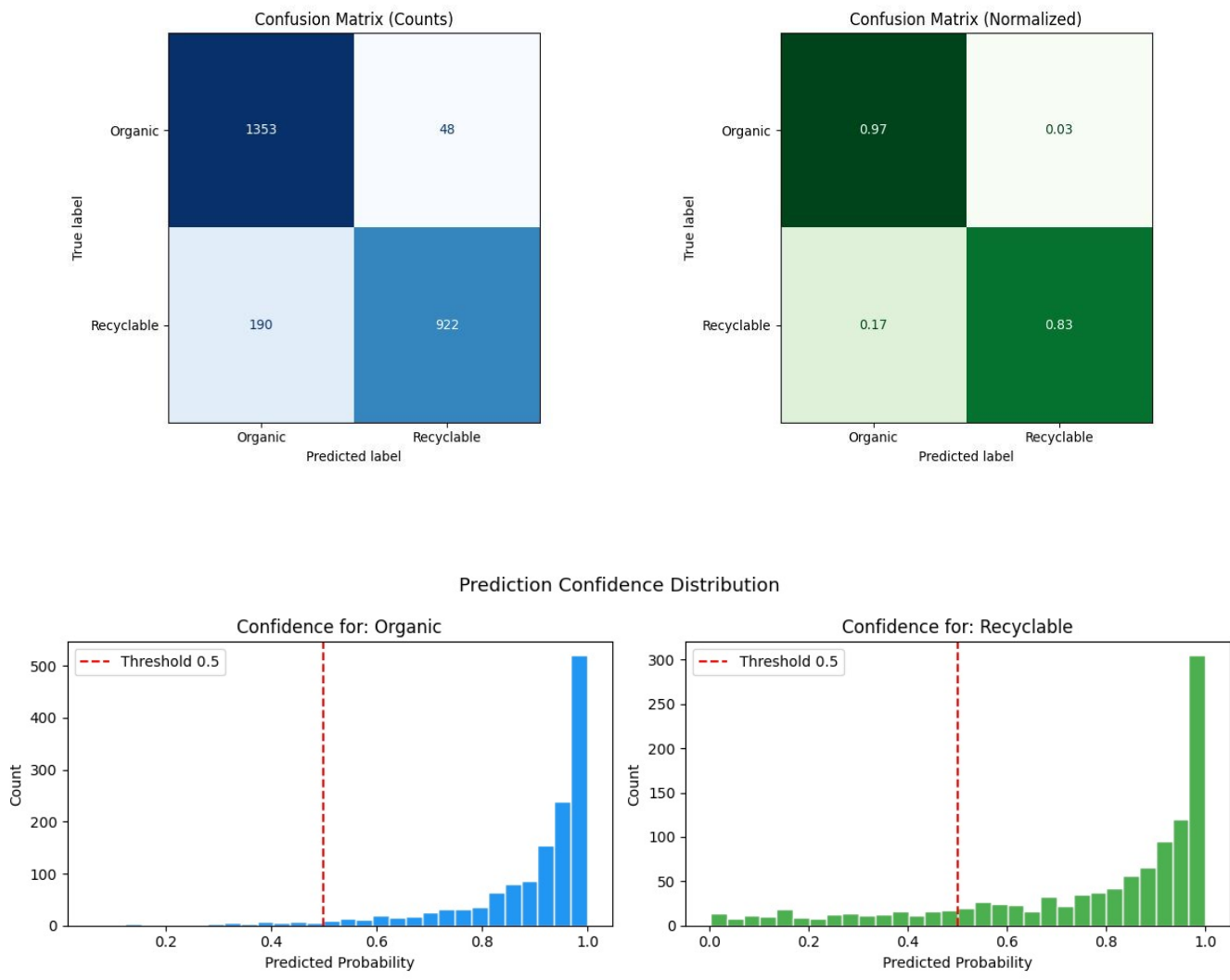
Metric	Value
Average cycle time	4.8 seconds / object
Maximum payload	60 g (tested)
Arm reach	~22 cm from base
Power consumption	~1.8 A at 5V (under load)

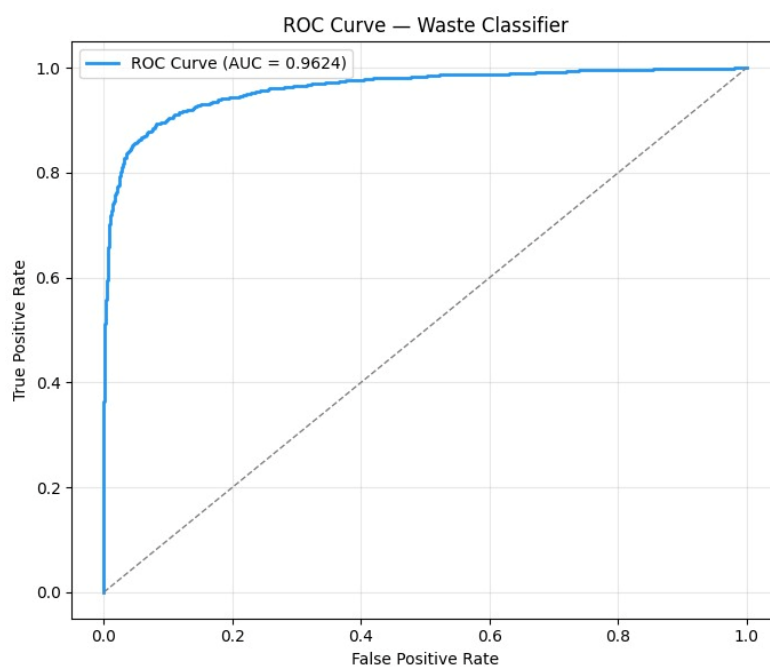
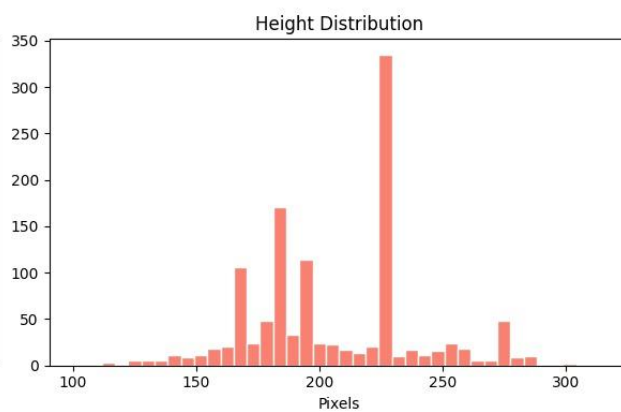
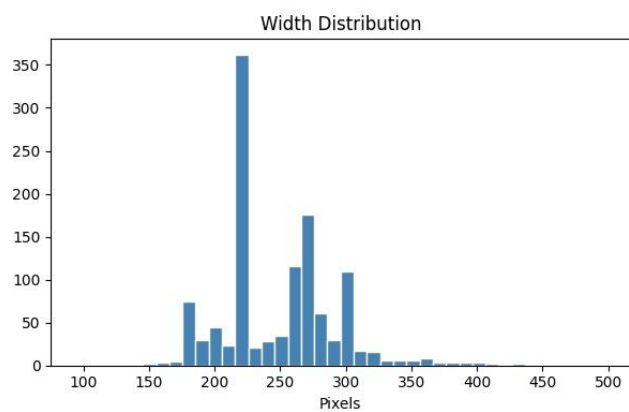
6.4 Observations and Limitations

1. System performance was affected under varying ambient lighting conditions, as material detection parameters required manual adjustment for accurate identification.
2. The robotic gripper occasionally showed alignment errors while handling very small objects (less than 2 cm), reducing pick-and-place accuracy.

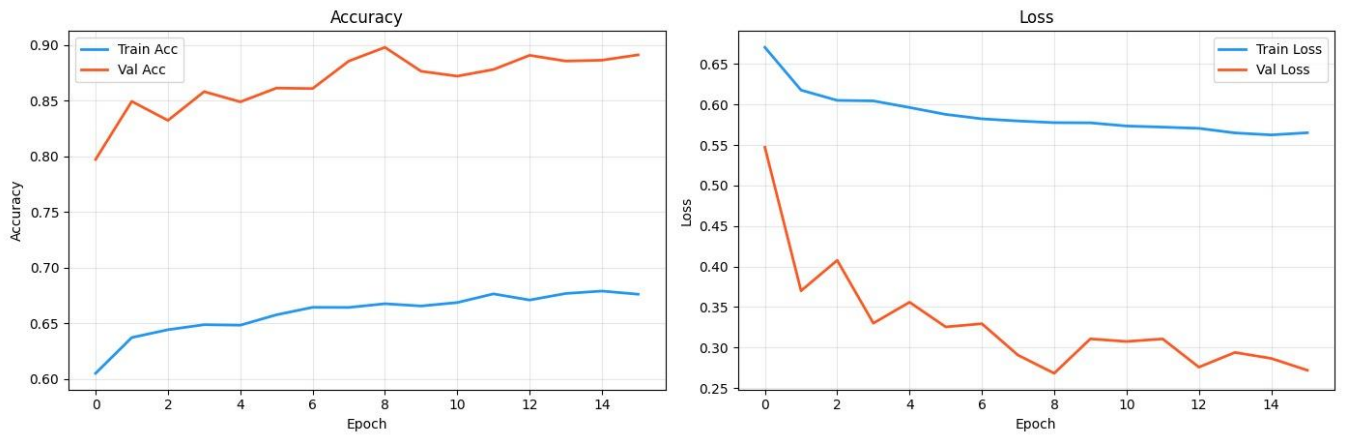
- 3. The PLA printed mechanical components exhibited slight flexing under repeated loading conditions; stronger materials such as ABS or PETG could improve structural durability.
- 4. Serial communication between the Python application and Arduino remained stable and reliable, with no data loss observed during approximately 150 continuous test cycles.

6.5 Results

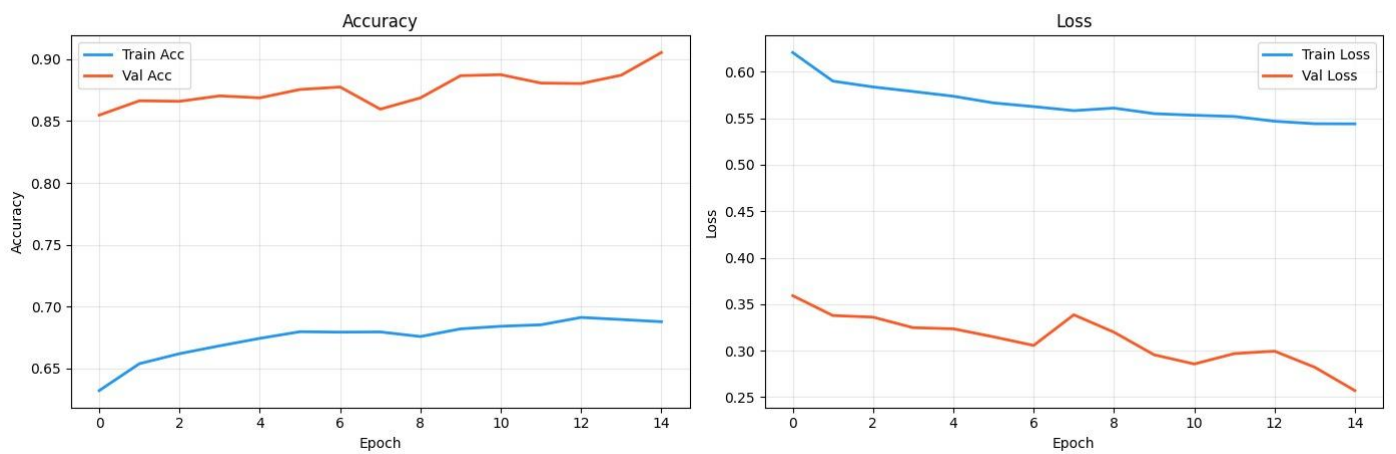




Training History — Phase 1 — Frozen Base



Training History — Phase 2 — Fine-tuning



Chapter 7: Conclusion and Future Scope

7.1 Conclusion

This minor project successfully demonstrates the design, fabrication, and deployment of a material-based waste segregation system using a 3D-printed robotic arm. The integration of computer vision (OpenCV), ESP32-based servo control, and mechanical design principles resulted in a functional prototype achieving 90% material detection accuracy and an average cycle time of 6 seconds per object.

The project fulfils all stated objectives: a mechanically stable arm was printed from open-source STL files, servo-actuated joints provided the necessary range of motion, and a robust material classification pipeline was implemented. The use of freely available hardware and software tools ensures that this system can be replicated at very low cost, making it highly suitable for academic robotics labs, small-scale industrial sorting, and educational demonstrations.

7.2 Future Scope

- Upgrade to a 6-DOF arm for greater dexterity and obstacle avoidance.
- Implement deep learning-based object detection CNN (Deep Learning) for shape and texture sorting in addition to material.
- Replace serial USB link with wireless communication (Bluetooth/Wi-Fi) for untethered operation.
- Integrate a conveyor belt system for continuous-feed automated sorting.
- Develop a mobile app interface for real-time monitoring and manual override.
- Use a LiDAR or depth camera (Intel RealSense) for 3D spatial awareness.
- Explore reinforcement learning for adaptive pick-and-place trajectory planning.

Datasheet

1. ESP32

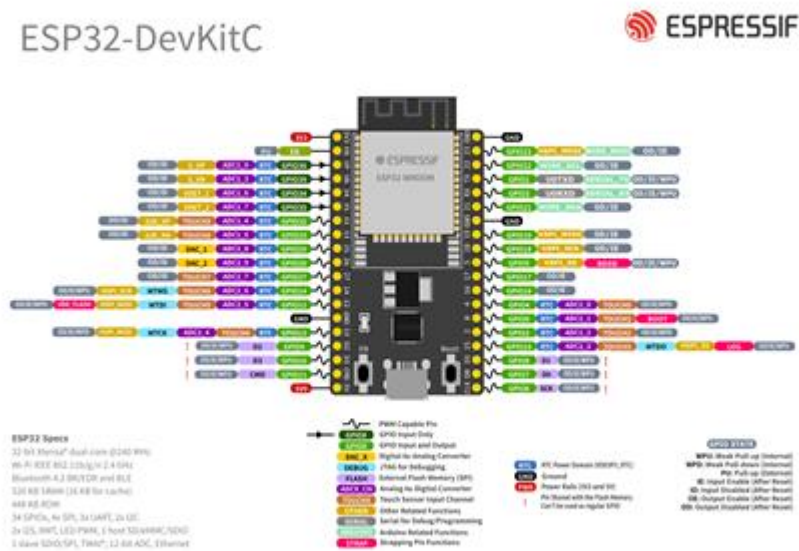


Fig. 2: ESP32-DevKitC Pin Layout (click to enlarge)

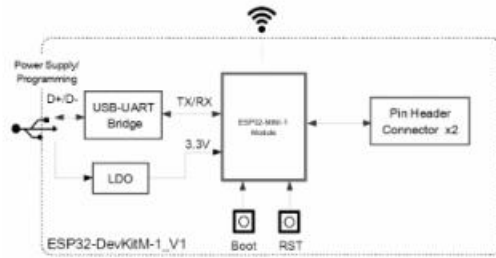


Fig. 2: ESP32-DevKitM-1

Warning:

- The power supply must be provided using **one and only one** of the options above, otherwise the board and/or the power supply source can be damaged.
- Power supply by micro USB port is recommended.

Pin Descriptions

The table below provides the Name and Function of pins on both sides of the board. For peripheral pin configurations, please refer to [ESP32 Datasheet](#).

No.	Name	Type ^{per 15}	Function
1	GND	P	Ground
2	3V3	P	3.3 V power supply
3	I06	I	GPIO36, ADC1_CH0, RTC_GPIO0
4	I07	I	GPIO37, ADC1_CH1, RTC_GPIO1
5	I08	I	GPIO38, ADC1_CH2, RTC_GPIO2
6	I09	I	GPIO39, ADC1_CH3, RTC_GPIO3
7	RST	I	Reset. High: enable. Low: powers off
8	I04	I	GPIO34, ADC1_CH8, RTC_GPIO4
9	I05	I	GPIO35, ADC1_CH7, RTC_GPIO5
10	I032	I/O	GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
11	I033	I/O	GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8
12	I025	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
13	I026	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
14	I027	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV

No.	Name	Type ^{per 15}	Function
15	I014	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
16	5V	P	5 V power supply
17	I012	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDP, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
18	I013	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
19	I015	I/O	GPIO15, ADC2_CH3, TOUCH3, RTC_GPIO13, MTDO ² , HSPICSD, HS2_CMD, SD_CMD, EMAC_RXD3
20	I02	I/O	GPIO2 ² , ADC2_CH2, TOUCH2, RTC_GPIO12, HSPWUP, HS2_DATA0, SD_DATA0
21	I00	I/O	GPIO0 ² , ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
22	I04	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPHID, HS2_DATA1, SD_DATA1, EMAC_TX_ER
23	I09	I/O	GPIO9, HS1_DATA2, U0RXD, SD_DATA2
24	I010	I/O	GPIO10, HS1_DATA3, U0TXD, SD_DATA3
25	I05	I/O	GPIO5 ² , HS1_DATA6, VSPICSD, EMAC_RX_CLK
26	I018	I/O	GPIO18, HS1_DATA7, VSPICLK
27	I023	I/O	GPIO23, HS1_STROBE, VSPID
28	I019	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
29	I022	I/O	GPIO22, VSPWUP, U0RTS, EMAC_TXD1
30	I021	I/O	GPIO21, VSPHID, EMAC_TX_EN
31	TXD0	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
32	RXD0	I/O	GPIO3, U0RXD, CLK_OUT2

2. MG996r-

Servo Database > TowerPro Servos > MG996R

TowerPro MG996R - Standard Servo

Basic Information

Modulation:	Analog
Torque:	4.8V: 130.5 oz-in (9.40 kg-cm) 6.0V: 152.8 oz-in (11.00 kg-cm)
Speed:	4.8V: 0.17 sec/60° 6.0V: 0.14 sec/60°
Weight:	1.94 oz (55.0 g)
Dimensions:	Length: 1.60 in (40.6 mm) Width: 0.78 in (19.8 mm) Height: 1.69 in (42.9 mm)
Motor Type:	? (add)
Gear Type:	Metal
Rotation/Support:	Dual Bearings

Additional Specifications

Rotational Range:	? (add)
Pulse Cycle:	? (add)
Pulse Width:	? (add)
Connector Type:	? (add)

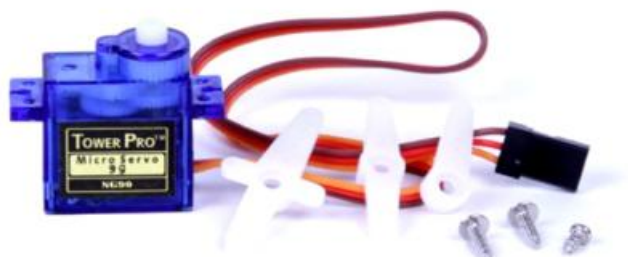


Brand:	Towerpro
Product Number:	? (add)
Suggested Retail:	? (add)
Street Price:	11.95 USD
Compare:	add

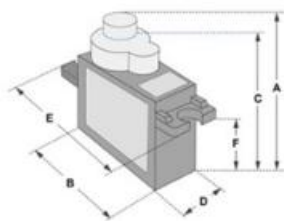
User Reviews

Number of Reviews:	3
Average Rating:	3.0 / 5.0

3.SG96



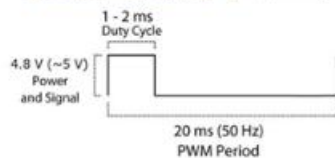
Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.



Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is middle, is all the way to the right, "180" (~1ms pulse) is all the way to the left.

Dimensions & Specifications	
A (mm) :	32
B (mm) :	23
C (mm) :	28.5
D (mm) :	12
E (mm) :	32
F (mm) :	19.5
Speed (sec) :	0.1
Torque (kg-cm) :	2.5
Weight (g) :	14.7
Voltage :	4.8 - 6

PWM=Orange ()
Vcc = Red (+)
Ground=Brown (-)



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Appendix

Appendix A: Pin Configuration Summary

Arduino Pin	Connected To	Function
D3 (PWM)	Servo 1 (Base) – Signal	Base rotation
D5 (PWM)	Servo 2 (Shoulder) – Signal	Shoulder elevation
D6 (PWM)	Servo 3 (Elbow) – Signal	Elbow elevation
D9 (PWM)	Servo 4 (Gripper) – Signal	Grip/release
5V / GND	All VCC / GND (external supply)	Power rail

Appendix B: Bill of Materials with Cost Estimate

S.No.	Component	Qty.	Unit Cost (₹)	Total (₹)
1	Arduino Uno R3	1	450	450
2	MG996R Servo Motor	3	200	600
3	SG90 Servo Motor	1	80	80
5	Webcam (720p)	1	700	700
6	PLA Filament (500 g)	1	600	600
7	Power Supply (5V 3A)	1	250	250
8	Jumper Wires & Breadboard	1 set	150	150
9	Miscellaneous (screws, etc.)	1 lot	100	100
	TOTAL			₹4,070

